

Morgan J. Henderson

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RELEVANT SKILLS

Programming Languages

Rust, Python, C++, MATLAB, C#, SQL, HTML, CSS, JavaScript, TypeScript, R

Machine Learning

Generative & recommender models, machine vision, reinforcement learning, uncertainty quantification, surrogate modeling, controls automation

Mathematics & Computer Science

Kalman & other Bayesian filters, optimization, parallel programming, eigen-analysis, numerical integration, applied statistics

Physics & Engineering

Vibrational analysis & lattice dynamics, magnetostatics, finite elements, RF modeling & simulation, experimental controls software, molecular dynamics, fire behavior modeling

EMPLOYMENT

AI/ML Engineer – Ladriss Technologies

February 2025 – February 2026

Languages: *Rust, Python, SQL, C++*

Skills: *Machine learning, MLOps, data pipelines, fire behavior modeling*

At Ladriss, I developed a flexible and highly general ML framework capable of meeting the company's growing model development and deployment needs. The framework, built on PyTorch, provided an API for building, training, loading, and deploying popular classes of deep-learning networks for use in the company's Rust-based backend services. I also developed (from scratch) and deployed a commercial-grade fire behavior simulation library and backend service to replace a vendor-provided service.

Associate Scientist & Scientist – RadiaSoft LLC

June 2022 – October 2024, October 2024 – February 2025

Languages: *Python, C++*

Skills: *Machine learning, MLOps, data pipelines, high energy and RF physics*

At RadiaSoft, I developed and deployed a variety of ML models for applications in high energy physics research. This included machine vision models (both standard CNNs and UNets) for automating neutron scattering experiments, LSTM-based variational autoencoders for sequence reconstruction and noise reduction, generative CNN autoencoders for predicting acceleration profiles in structure-based wakefield accelerators, and more. I also built a framework for streamlining model development and deployment and simulation software for RF systems used in particle accelerators.

Research & Teaching Assistant – University of Colorado Boulder

August 2018 – May 2022

Languages: *C++, Python, Fortran, MATLAB, R*

Skills: *Materials engineering, molecular dynamics, data pipelines, machine learning*

As a graduate research assistant, I conducted detailed research on the subject of thermal transport via wave phenomena at the nanoscale in semiconductor materials. My day-to-day tasks included performing large scale simulations and intensive calculations on atomically resolved nanostructures and analyzing the resulting data, as well as performing in-depth literature reviews into related areas of research.

Analyst / Programmer I & II – State of Alaska

November 2017 – August 2018

Languages: *C#, JavaScript/TypeScript, SQL, HTML, CSS*

Skills: *Webapp development, project management, website maintenance*

As an Analyst/Programmer for the State of Alaska, Department of Education & Early Development, I was responsible for designing, constructing, testing, and deploying web applications to automate both internal and external departmental processes. As the department Webmaster, I was also tasked with updating and improving the department website and handling internal change requests.

Visiting Researcher – NASA Frontier Development Lab

June 2017 – August 2017

Languages: *Python*

Skills: *Machine learning, solar physics, machine vision*

At the NASA Frontier Development Lab, I was responsible for providing physical context and insight regarding planetary science problems being addressed by ML experts in the program. I also provided programming support to other planetary scientists and spent a significant amount of time learning about and interfacing with the algorithms employed by other mixed teams of ML and planetary science experts.

EDUCATION

M.S. Aerospace Engineering Sciences – University of Colorado Boulder

August 2018 – May 2022

Primary emphasis in Fluids, Structures & Materials, secondary emphasis in Automatic Controls

B.A Physics – University of Montana

August 2012 – May 2017

High Honors, focus in Astronomy

B.A Mathematics – University of Montana

August 2012 – May 2017

High Honors, focus in Applied Mathematics

CERTIFICATES

Machine Learning – Stanford University Online

May 2022 – June 2022

Engineering Management in the Aerospace Industry – University of Colorado Boulder

May 2022 – June 2022

WEB & MEDIA

Website

Visit [my personal website](#), which I built from scratch and host through GitHub

Open-Source Projects

Browse [my open-source projects](#), hosted on Codeberg

LinkedIn

Check out [my LinkedIn page](#), and feel free to connect with me there

Videos

[RadiaSoft webinar, hysteresis modeling \(virtual, 2022\)](#)

[NASA FDL final talks \(Intel Auditorium, Santa Clara CA, 2017\)](#)

[Senior thesis talk \(University of Montana, Missoula MT, 2017\)](#)

[SETI Institute REU lightning talks \(Microsoft Auditorium, Mountain View CA, 2016\)](#)